Ch 9.2: Support Vector Classifier

Lecture 26 - CMSE 381

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Fri, Nov 18, 2022

Announcements

Last time:

- 9.1 Maximal Margin Classifier

This lecture:

- 9.2 Support Vector Classifier

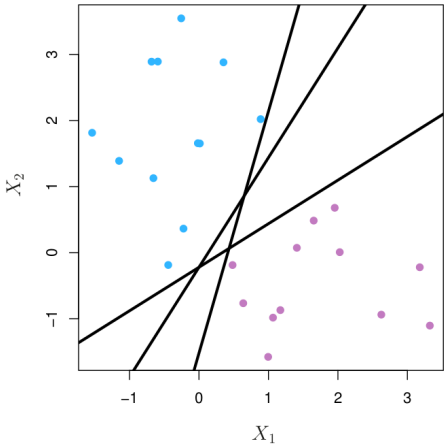
Announcements:

- HW #8 due tonight

Section 1

Last time

Separating Hyperplane



Require that for every data point:

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$

Equivalently

Require that for every data point

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) > 0$$

Separating hyperplane becomes a classifier

$f(2,3) = \beta_0 + \beta_1(2) + \beta_2(3)$

If you have a separating hyperplane:

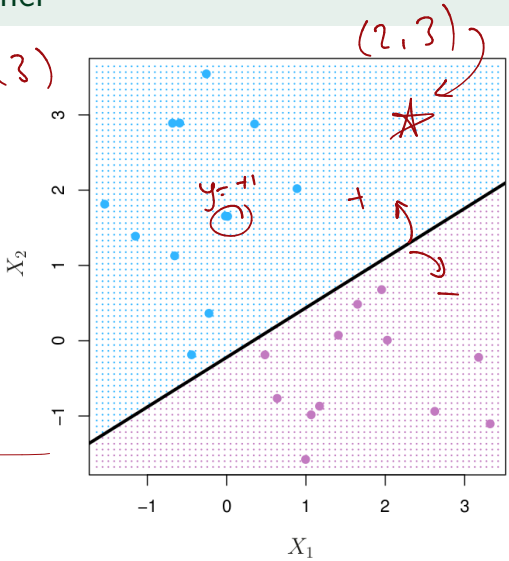
- Check

$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

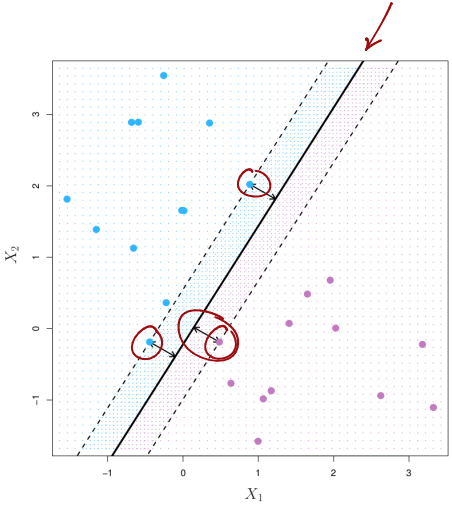
- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$

$1 + 2x_1 + 3x_2 = 0$

$-1 - 2x_1 - 3x_2 = 0$



Maximal margin classifier



- For a hyperplane, the *margin* is the smallest distance from any data point to the hyperplane.
- Observations that are closest are called *support vectors*.
- The *maximal margin hyperplane* is the hyperplane with the largest margin
- The classifier built from this hyperplane is the *maximal margin classifier*.

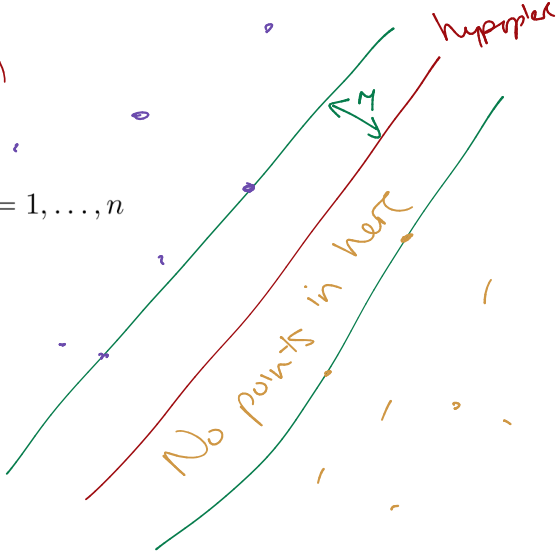
Mathematical Formulation

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

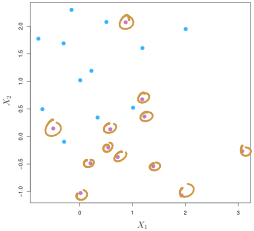
(unit normal)

$$\underbrace{y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip})}_{\text{distance from point } i \text{ to the hyperplane}} \geq M \quad \forall\, i = 1, \ldots, n$$

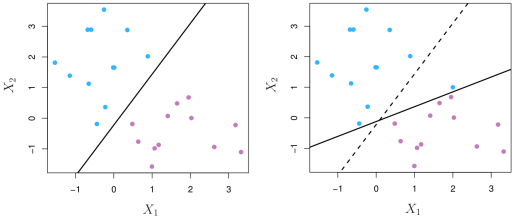


Problems

Might be no separating hyperplane



Sensitivity to new points



Section 2

Support Vector Classifier

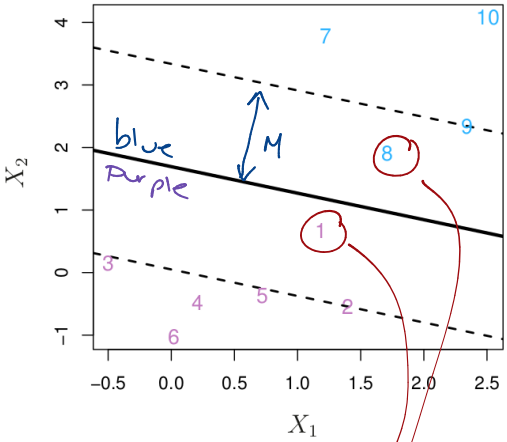
Basic idea

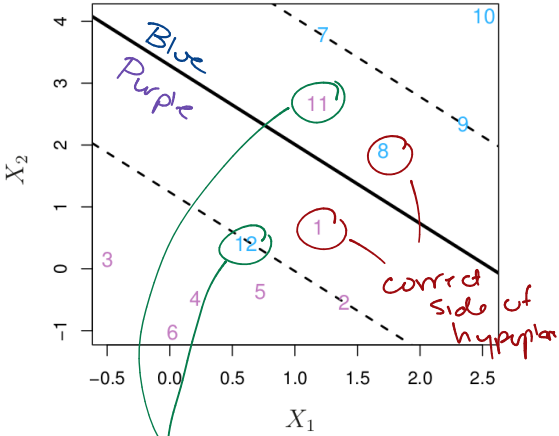
- Be ok with a not perfect classifier
- Aim for greater robustness to individual observations (Problem 2 on prev slide)
- Better classification of <u>most</u> of the data points

Support Vector Classifier (SVC)

or Soft Margin Classifier

Soft margin





Points on the wrong side of margin

Wrong side of the hyperplane (misclassified)

Mathematical Formulation of SVC

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,M}{\text{maximize}} \quad M$$

width of margin

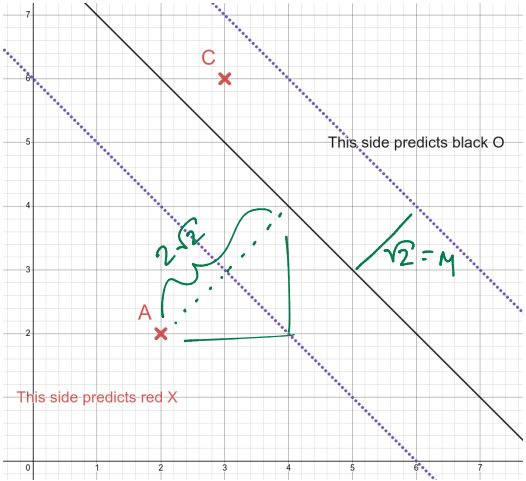
$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

Find positive ε's that will satisfy this

Fix $M = \sqrt{2}$ $\quad \overbrace{y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip})} \geq M(1 - \varepsilon_i)$



$A = (2,2)$ dist to hyperplane

$= 2\sqrt{2}$ = left side

$\geq M(1-\varepsilon_i)$

$= \sqrt{2}(1-\varepsilon_i)$

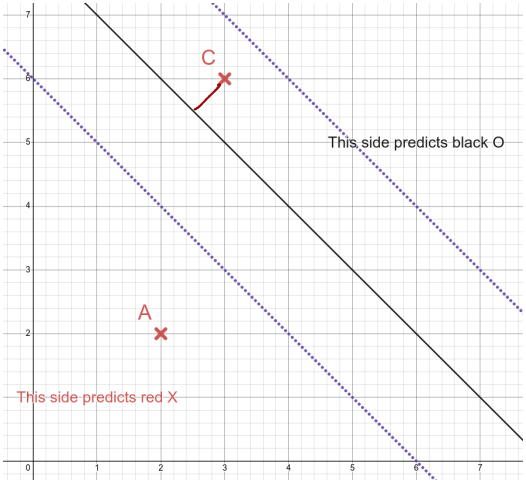
$2\sqrt{2} \overset{?}{\geq} \sqrt{2}(1-\varepsilon_i)$

$2 \geq 1-\varepsilon_i$

If $\varepsilon = 0 \rightsquigarrow$ this is true!

Find positive ε's that will satisfy this

Fix $M = \sqrt{2}$ $\quad y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \varepsilon_i)$



$C = (3,6)$ $\quad$ dist $= \frac{\sqrt{2}}{2}$

left side $= -\frac{\sqrt{2}}{2}$

Want ε_i

$$-\frac{\sqrt{2}}{2} \geq \sqrt{2}(1 - \varepsilon_i)$$

$$-\frac{1}{2} \geq 1 - \varepsilon_i$$

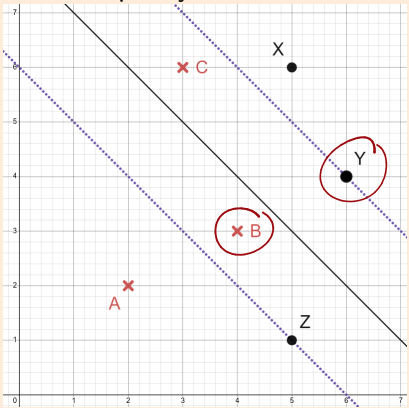
$$\varepsilon_i \geq 1.5$$

$$\varepsilon_i = 1.5$$

What is ε?

Fix $M = \sqrt{2}$ $\qquad y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \varepsilon_i)$

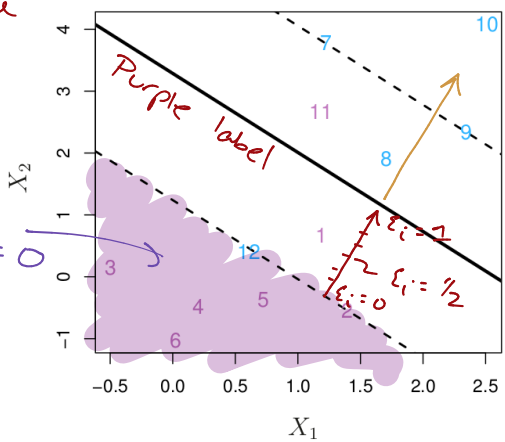
Fill in the table so that the inequality is satisfied.



Point	Left Side	ε_i	$M(1-\varepsilon_i)$
A	$2\sqrt{2}$	0	$\sqrt{2}$
B	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	$\sqrt{2}(1-\frac{1}{2}) = \frac{\sqrt{2}}{2}$
C	$-\frac{\sqrt{2}}{2}$	1.5	$-\frac{\sqrt{2}}{2}$
X	$\frac{3\sqrt{2}}{2}$	0	$\sqrt{2}$
Y	$\sqrt{2}$	0	$\sqrt{2}$
Z	$-\sqrt{2}$	2	$\sqrt{2}(1-2)$ $\sqrt{2}(-1) = -\sqrt{2}$

What is ε?

- pt on correct side of hyperplane
  & correct side of margin
  $\varepsilon_i = 0$
- Correct side hyperplane
  but wrong side of margin
  $0 \le \varepsilon_i \le 1$
- wrong side of hyperplane
  $\varepsilon_i \ge 1$



What is C?

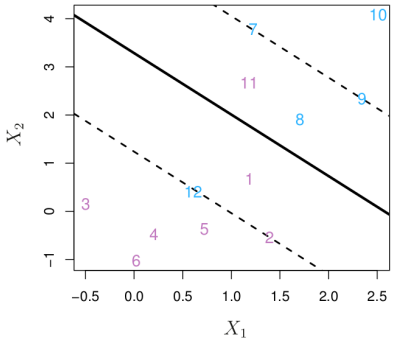
$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

↳ Budget for Violations

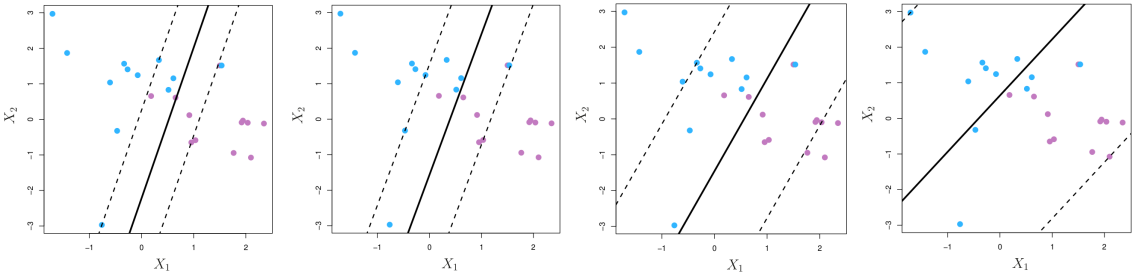


Prev example $\sum \epsilon_i \leq 4$

If $C = 0$

↳ Maximal margin classifier

Examples messing with C



Increasing $C \rightarrow$

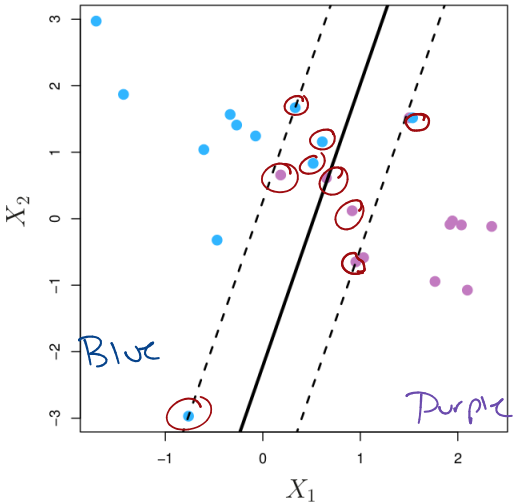
narrower margins

$\Uparrow$

"Book's C"

Wider margins allowed

What affects the hyperplane?



Support vectors
pts on or violate
the margin

Coding

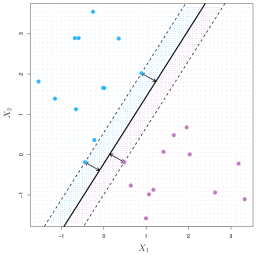
TL;DR

Maximal Margin Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \; M$$

$$\text{subject to} \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \;\; \forall\, i = 1, \ldots, n$$



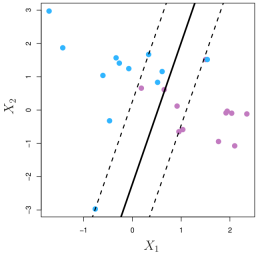
Support Vector Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\, M}{\text{maximize}} \quad M$$

$$\text{subject to} \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \;\; \sum_{i=1}^{n} \epsilon_i \leq C,$$



Next time

20	F	Nov 4	Polynomial & Step Functions.	7.1,7.2	
21	M	Nov 7	Step Functions	7.2	
22	W	Nov 9	Basis functions, Regression Splines	7.3,7.4	
23	F	Nov 11	Decision Trees	8.1	HW #7 Due
24	M	Nov 14	Random Forests	8.2.1, 8.2.2	
25	W	Nov 16	Maximal Margin Classifier	9.1	
26	F	Nov 18	SVC	9.2	HW #8 Due
27	M	Nov 21	SVM	9.3, 9.4, 9.5	
28	W	Nov 23	Extended virtual office hours		
	F	Nov 25	No class - Thanksgiving		
29	M	Nov 28	Single layer NN	10.1	HW #9 Due
30	W	Nov 30	Multi Layer NN	10.2	
31	F	Dec 2	CNN	10.3	
32	M	Dec 5	Unsupervised Learning & Clustering	12.1, 12.4	HW #10 Due
	W	Dec 7	Review		
	F	Dec 9	Midterm #3	Bring your cheat sheet and a non-internet-connected calculator	